MMDU MULLANA OFFICIAL SYLLABUS • BELE-001 • UNIT 2

Unit 2: Transformers & Electrical Machines (The Magic Gearbox)

Subject: Basic Electrical & Electronics Engineering (BELE-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

Transformers explained as magnetic gearboxes! Step-up vs Step-down, EMF equation derivation, Core vs Copper losses, OC/SC tests explained simply, DC Motor vs Generator working, and Fleming's Left & Right Hand Rules.

1. Single-Phase Transformer Working & EMF Equation

1. Electromagnetic Principles of the Single-Phase Transformer

A **transformer** is a static electromagnetic device that transfers electrical energy between two or more circuits via mutual magnetic induction at constant frequency.

Working Principle & Faraday's Law:

1. When sinusoidal voltage $v_1(t) = V_{m1}\sin(\omega t)$ is connected across primary winding of N_1 turns, an alternating magnetizing current flows, producing sinusoidal core flux:

$$\phi(t) = \phi_m \sin(\omega t)[\text{Weber, Wb}]$$

2. This common mutual flux links both primary and secondary windings through the high-permeability laminated ferromagnetic core.

3. By Faraday's Law of Induction, counter-EMFs are induced in both windings:

$$e_1 = -N_1\frac{d\phi}{dt}, e_2 = -N_2\frac{d\phi}{dt}$$

2. Rigorous Derivation of the Transformer EMF Equation

Substitute sinusoidal flux $\phi(t) = \phi_m \sin(\omega t)$:

$$e_1 = -N_1\frac{d}{dt}[\phi_m \sin(\omega t)] = -N_1\omega\phi_m \cos(\omega t)$$

Using the trigonometric identity $-\cos\theta = \sin(\theta - 90°)$:

$$e_1 = N_1(2\pi f)\phi_m \sin(\omega t - 90°)$$

The maximum (peak) induced EMF is:

$$E_{m1} = 2\pi f N_1 \phi_m$$

The Root Mean Square (RMS) value is:

$$E_1 = \frac{E_{m1}}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} f N_1 \phi_m = \sqrt{2}\pi f N_1 \phi_m$$

Since $\sqrt{2}\pi \approx 1.4142 \times 3.1416 \approx 4.44288$:

$$\mathbf{E_1 = 4.44 \cdot f \cdot N_1 \cdot \phi_m}$$

Similarly, for the secondary winding:

$$\mathbf{E_2 = 4.44 \cdot f \cdot N_2 \cdot \phi_m}$$

EMF per Turn (Core Invariant):

Dividing by the turn count yields the fundamental invariant of transformer design:

$$\mathbf{\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44 \cdot f \cdot \phi_m = 4.44 \cdot f \cdot B_m \cdot A_i}$$

where B_m is maximum core flux density (Tesla) and A_i is net iron cross-sectional area (m^2).

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3. Transformation Ratio (K) & Impedance Reflection

$$\mathbf{K = \frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}}$$

- $K > 1 \Rightarrow$ **Step-Up Transformer** $(V_2 > V_1, I_2 < I_1)$.
- $K < 1 \Rightarrow$ **Step-Down Transformer** $(V_2 < V_1, I_2 > I_1)$.

Resistance and Reactance Referral Rules:

- Transferring secondary impedance to primary side:

$$\mathbf{R_2' = \frac{R_2}{K^2}, x_2' = \frac{x_2}{K^2}, Z_{eq1} = R_1 + R_2' + j(x_1 + x_2')}$$

- Transferring primary impedance to secondary side:

$$\mathbf{R_1'' = K^2 R_1, x_1'' = K^2 x_1, Z_{eq2} = R_2 + R_1'' + j(x_2 + x_1'')}$$

Problem:
A single-phase transformer has 500 primary turns and 50 secondary turns. If the primary is connected to a 230 V AC source, calculate the secondary voltage V_2 and transformation ratio $K.

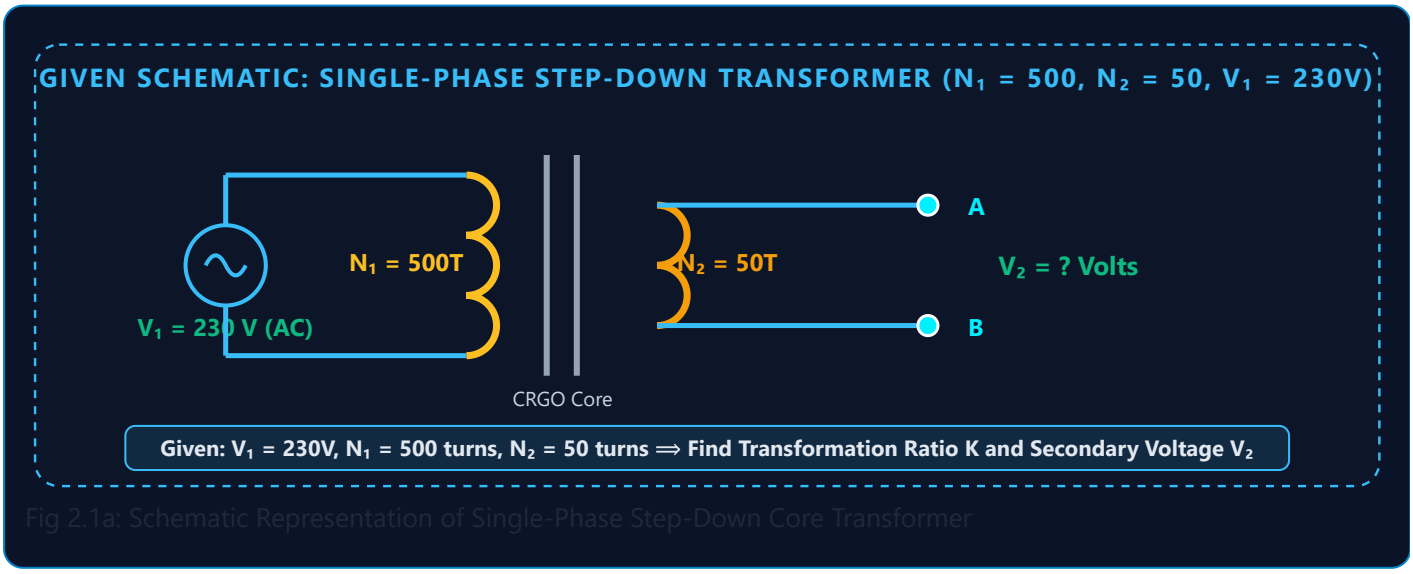


Fig 2.1a: Schematic Representation of Single-Phase Step-Down Core Transformer

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Step 1 🚀: Transformation Ratio K

$$K = \frac{N_2}{N_1} = \frac{50}{500} = \mathbf{0.10}(\text{Step-down transformer})$$

Step 2 ⚡: Secondary Voltage

$V_2 = K \cdot V_1 = 0.10 \times 230\ \text{V} = \mathbf{23.0}$ Volts

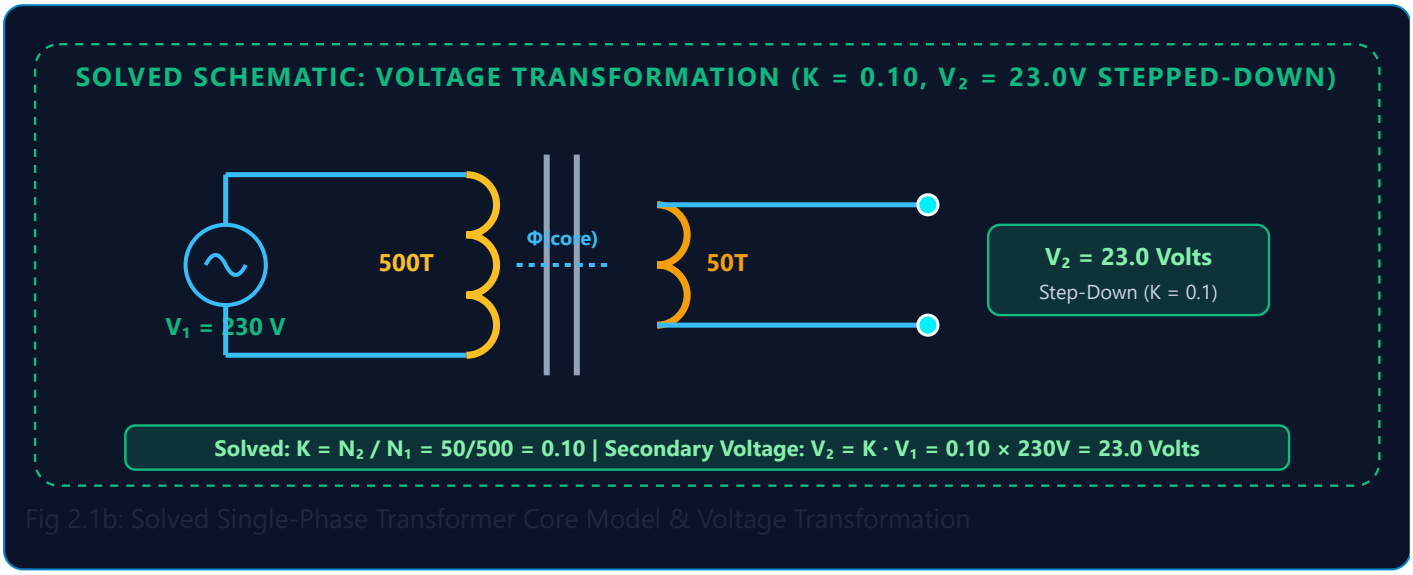


Fig 2.1b: Solved Single-Phase Transformer Core Model & Voltage Transformation

🎯 Transformation Ratio $K = 0.10$ | Secondary Voltage $V_2 = 23.0$ V.

University Question (8 Marks):

A 2200 / 220 V, 50 Hz single-phase step-down transformer has 60 turns on the secondary winding.

Calculate: (1) Primary winding turns N_1, (2) Peak magnetic flux in the core φ_m, (3) Full-load primary current when secondary delivers 20 A.

Step 1 🚀: Primary turns N_1

$(N_1 / N_2) = (V_1 / V_2) \Rightarrow (N_1 / 60) = \frac{2200}{220} = 10 \Rightarrow N_1 = 600$, urns

Step 2 ⚡: Peak magnetic flux φ_m via EMF equation

$E_2 = 4.44 \cdot f \cdot N_2 \cdot \varphi_m \Rightarrow 220 = 4.44 \times 50 \times 60 \times \varphi_m$

$220 = 13320 \times \varphi_m \Rightarrow \varphi_m = \frac{220}{13320} = 0.0165\,\text{Wb} = 16.5\,\text{(mWb)}$

Step 3 🎯: Primary current I_1

$I_1 = I_2 \cdot \left((N_2 / N_1)\right) = 20 \times \left(\frac{60}{600}\right) = 2.0\,\text{Amperes}$

🎯 Primary Turns $N_1 = 600$ | Peak Flux $\phi_m = 16.5$ mWb | Primary Current $I_1 = 2.0$ A.

University Question (10 Marks):

A 3300 / 230 V, 50 Hz single-phase transformer has a maximum flux density $B_m = 1.2$ Tesla in the core. The induced EMF per turn is 8.0 Volts.

Calculate: (1) Number of primary and secondary turns, (2) Net cross-sectional area of the core A_i.

Step 1 🚀: Primary and Secondary Turns

EMF per turn = (E_1 / N_1) = (E_2 / N_2) = 8.0 V/turn

N_1 = (E_1 / EMF/turn) = \frac{3300}{8} = 412.5 ⇒ 413, urns

N_2 = (E_2 / EMF/turn) = \frac{230}{8} = 28.75 ⇒ 29, urns

Step 2 ⚡: Peak Magnetic Flux φ_m

EMF per turn = 4.44 · f · φ_m ⇒ 8.0 = 4.44 × 50 × φ_m

φ_m = (8.0 / 222) = 0.03604,Weber = 36.04,\text{mWb}

Step 3 🎯: Core Cross-Sectional Area A_i

φ_m = B_m × A_i ⇒ A_i = (φ_m / B_m) = (0.03604 / 1.2) = 0.03003,m^2 = 300.3~\text{cm^2}

🎯 $N_1 = 413$ turns, $N_2 = 29$ turns | $\phi_m = 36.04$ mWb | Core Area $A_i = 300.3$ cm^2.

2. Transformer Losses, Efficiency & The OC/SC Tests

1. In-Depth Analysis of Transformer Losses & Efficiency

A. Core Losses / Iron Losses (P_i — Invariant with Load):

Occur inside the magnetic core due to cyclic reversal of magnetization:

1. **Hysteresis Loss (P_h):** Energy dissipated aligning molecular magnetic dipoles:

$$\mathbf{P_h} = \eta_h \cdot \mathbf{B_m^{1.6}} \cdot \mathbf{f} \cdot \mathbf{V_{core}} [\text{Steinmetz Equation}]$$

- Minimized by using **Cold-Rolled Grain-Oriented (CRGO) Silicon Steel** with a narrow hysteresis loop.

2. **Eddy Current Loss (P_e):** Joule heating from induced circular eddy currents in the conductive core:

$$\mathbf{P_e} = \mathbf{K_e} \cdot \mathbf{B_m^2} \cdot \mathbf{f^2} \cdot \mathbf{t^2} \cdot \mathbf{V_{core}}$$

- Minimized by slicing the solid core into **thin insulated laminations** of thickness $t \approx 0.25 - 0.35$ mm varnished on both sides.

B. Copper Losses (P_{cu} — Variable with Load):

Joule heating in the electrical windings:

$$\mathbf{P_{cu}} = \mathbf{I_1^2 R_1} + \mathbf{I_2^2 R_2} = \mathbf{I_2^2 R_{eq2}} = \mathbf{x^2 P_{cu\ FL}}$$

where $x = \frac{I_2}{I_{2\,\mathrm{FL}}} = \frac{\text{Load kVA}}{\text{Rated kVA}}$ is the fractional loading factor.

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2. Efficiency Formulation & Maximum Efficiency Derivation

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} = \frac{\mathbf{x} \cdot \mathbf{S}_{\text{rated}} \cdot \cos\phi_2}{\mathbf{x} \cdot \mathbf{S}_{\text{rated}} \cdot \cos\phi_2 + \mathbf{P_i} + \mathbf{x^2 P_{cu\,FL}}}$$

Condition for Maximum Efficiency:

Differentiating η with respect to load current I_2 and setting to zero:

$$\frac{d\eta}{dI_2} = 0 \Rightarrow \mathbf{P_i} = \mathbf{x^2 P_{cu\,FL}} \iff \text{Constant Iron Loss} = \text{Variable Copper Loss}$$

- Fractional load at maximum efficiency:

$$\mathbf{x}_{\max} = \sqrt{\frac{\mathbf{P_i}}{\mathbf{P_{cu\,FL}}}}$$

- Load kVA for maximum efficiency:

$$\text{kVA}_{\max} = \text{Rated kVA} \times \sqrt{\frac{\mathbf{P_i}}{\mathbf{P_{cu\,FL}}}}$$

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3. Open Circuit (OC) and Short Circuit (SC) Test Formulations

Test Metric	Open Circuit (OC) Test	Short Circuit (SC) Test
Objective	Determine Core Loss P_i, R_0, and X_0	Determine Full-Load Copper Loss P_{cu}, R_{eq}, X_{eq}
Winding Energized	Low Voltage (LV) winding	High Voltage (HV) winding
Opposite Winding	High Voltage left open ($I_2 = 0$)	Low Voltage shorted with thick copper conductor
Excitation Level	Rated rated voltage V_1 at rated frequency f	Small reduced voltage (5 of rated)
Key Formulae	\cos\phi_0 = \frac{W_{oc}}{V_1 I_0} $I_c = I_0 \cos\phi_0 \Rightarrow R_0 = \frac{V_1}{I_c}$ I_m = I_0 \sin\phi_0 ⇒ X_0 = \frac{V_1}{I_m}	R_{eq} = \frac{W_{sc}}{I_{sc}^2} $Z_{eq} = \frac{V_{sc}}{I_{sc}}$ X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}

Problem:
A 10 kVA transformer has iron loss $P_i = 150$ W and full-load copper loss $P_{cu} = 250$ W. Calculate its full-load efficiency at unity power factor ($\cos\phi = 1.0$).

Step 1 🚀: Output Power

$$P_{\text{out}} = \text{kVA} \times 1000 \times \cos\phi = 10 \times 1000 \times 1.0 = \mathbf{10{,}000} \text{ Watts}$$

Step 2 ⚡: Total Losses

$$P_{\text{loss}} = P_i + P_{cu} = 150 + 250 = \mathbf{400} \text{ Watts}$$

Step 3 🎯: Efficiency Formula

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \times 100 = \frac{10000}{10000 + 400} \times 100 = \frac{10000}{10400} \times 100 = \mathbf{96.15\%}$$

🎯 Full-Load Efficiency: $\eta = \mathbf{96.15\%}$.

University Question (8 Marks):
A 25,kVA transformer has iron loss P_i = 200 W and full-load copper loss Pc_u = 400 W. Find:

1. Efficiency at half-load with 0.8 power factor lagging.

2. The load kVA at which maximum efficiency occurs.

Step 1 🚀: Half-Load (x = 0.5) Calculation

- Output power: P_{out} = (0.5)(25 × 10^3)(0.8) = 10{,000 W}
- Copper loss at half-load: Pc_u' = x^2 Pc_u = $(0.5)^2$ × 400 = 0.25 × 400 = 100 W
- Constant iron loss: P_i = 200 W
- Efficiency:

$\eta_{50}\%$ = \frac{10000}{10000 + 100 + 200} × 100 = \frac{10000}{10300} × 100 = 97.09%

Step 2 ⚡: Load kVA for Maximum Efficiency ($Pc_u = P_i$)

kVA_{max} = Rated kVA × √((P_i / Pc_u)) = 25 × √(\frac{200}{400}) = 25 × √(0.5) = 25 × 0.7071 = 17.68,kVA

🎯 Half-Load Efficiency = 97.09 | Maximum Efficiency occurs at **17.68** kVA.

University Question (12 Marks - University Exam):

The following test results were obtained on a 5,kVA, 200 / 400 V, 50 Hz single-phase transformer:

- **OC Test (LV side):** 200 V, 0.7 A, 60 W
- **SC Test (HV side):** 22 V, 12.5 A, 120 W

Find the equivalent circuit parameters referred to the LV side.

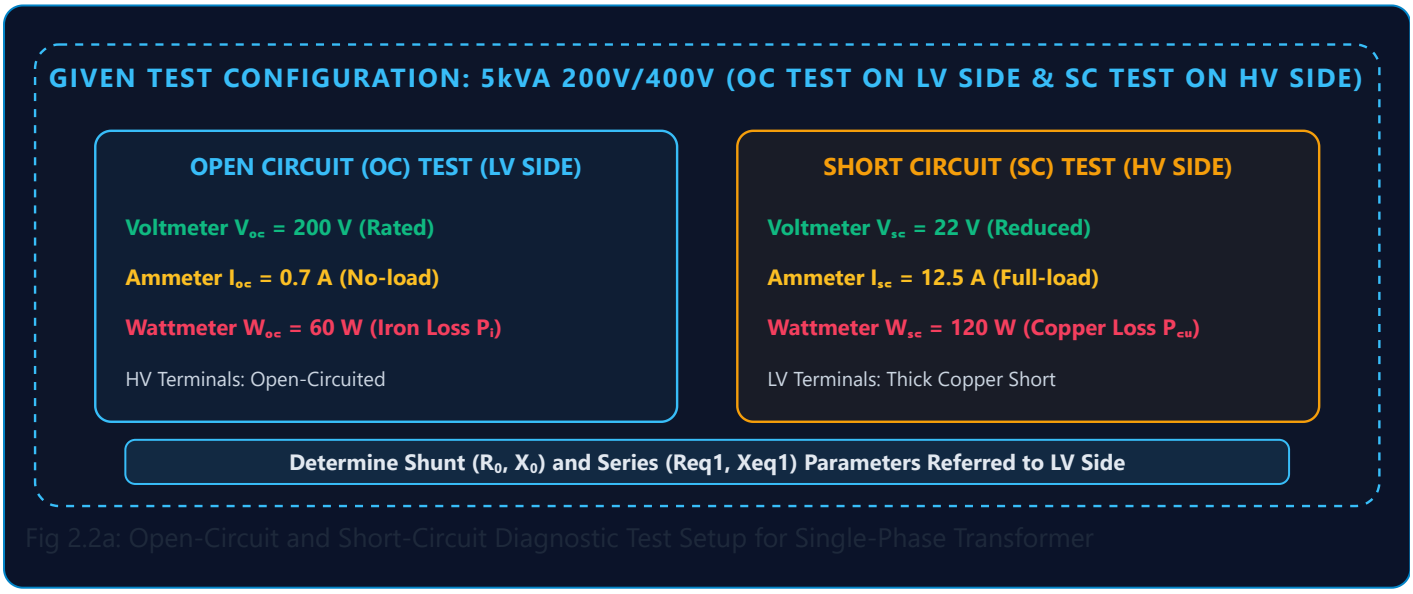


Fig 2.2a: Open-Circuit and Short-Circuit Diagnostic Test Setup for Single-Phase Transformer

Step 1 🚀: OC Test Parameters (Shunt Branch R_0, X_0)

$$\cos\phi_0 = \frac{W_{oc}}{V_{oc}I_{oc}} = \frac{60}{200 \times 0.7} = \frac{60}{140} = 0.4286$$

$$\sin\phi_0 = \sqrt{1-(0.4286)^2} = 0.9035$$

- Core loss current: $I_c = I_{oc}\cos\phi_0 = 0.7 \times 0.4286 = 0.30$ A
- Magnetizing current: $I_m = I_{oc}\sin\phi_0 = 0.7 \times 0.9035 = 0.632$ A

$$R_0 = \frac{V_{oc}}{I_c} = \frac{200}{0.30} = \mathbf{666.7\ \Omega}$$

$$X_0 = \frac{V_{oc}}{I_m} = \frac{200}{0.632} = \mathbf{316.5\ \Omega}$$

Step 2 ⚡: SC Test Parameters (Series Branch referred to HV side)

$$R_{eq2} = \frac{W_{sc}}{I_{sc}^2} = \frac{120}{(12.5)^2} = \frac{120}{156.25} = \mathbf{0.768\ \Omega}$$

$$Z_{eq2} = \frac{V_{sc}}{I_{sc}} = \frac{22}{12.5} = \mathbf{1.76\ \Omega}$$

$$X_{eq2} = \sqrt{Z_{eq2}^2 - R_{eq2}^2} = \sqrt{(1.76)^2-(0.768)^2} = \sqrt{3.098-0.590} = \mathbf{1.584\ \Omega}$$

Step 3 🎯: Refer Series Branch to LV Side ($K = 400/200 = 2$)

$$R_{eq1} = \frac{R_{eq2}}{K^2} = \frac{0.768}{4} = \mathbf{0.192\ \Omega}$$

$X_{eq1} = \frac{X_{eq2}}{K^2} = \frac{1.584}{4} = \mathbf{0.396\ \Omega}$

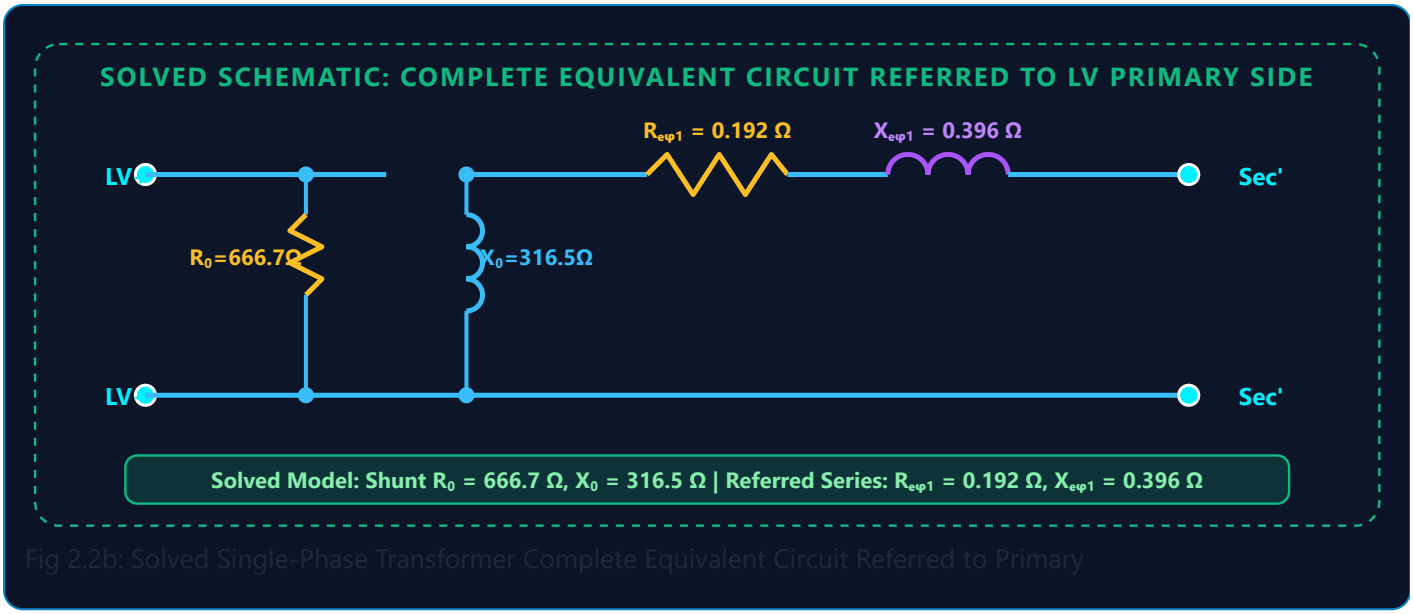


Fig 2.2b: Solved Single-Phase Transformer Complete Equivalent Circuit Referred to Primary

🎯 LV Referred Parameters: $R_0 = 666.7\ \Omega, X_0 = 316.5\ \Omega$ | $R_{eq1} = 0.192\ \Omega, X_{eq1} = 0.396\ \Omega$.

3. DC Motors & Generators (Fleming's Hand Rules)

1. DC Machines: Construction, Action & Hand Rules

A **Direct Current Machine** is an electromechanical energy converter operating bidirectionally:

- **DC Generator:** Mechanical shaft torque → DC electrical power via Faraday's Law.
- **DC Motor:** DC electrical power → Mechanical shaft torque via Lorentz Force Law.

Structural Components:

1. **Stator:** Yoke (magnetic frame), Field poles with pole shoes, Shunt/Series Field windings.
2. **Rotor (Armature):** Laminated cylindrical silicon-steel core, insulated armature conductors in slots.
3. **Commutator:** Split-ring copper wedge segments insulated with mica, acting as a mechanical rectifier/inverter.
4. **Brushes:** High-grade carbon blocks riding on the commutator to collect or inject current.

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2. Fleming's Hand Rules: Left Hand (Motor) vs Right Hand (Generator)

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FLEMING'S LEFT-HAND RULE (MOTORS)              FLEMING'S RIGHT-HAND RULE (GENERATORS)
          Thumb = Force (Motion)                         Thumb = Motion of Conductor
          Forefinger = Magnetic Field (N -> S)           Forefinger = Magnetic Field (N -> S)
          Middle Finger = Current                        Middle Finger = Induced Current
```

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3. Mathematical Derivations of DC Machine Equations

A. General Induced EMF Equation:

Let:

- P = Number of poles, ϕ = Magnetic flux per pole (Wb)
- Z = Total number of armature conductors, N = Rotational speed (rpm)
- A = Number of parallel paths ($A = 2$ for Wave winding; $A = P$ for Lap winding)

Flux cut by one conductor in 1 revolution: $d\phi = P\phi$.
Time for 1 revolution: $dt = \frac{60}{N}$ seconds.
EMF induced in one conductor: $e = \frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{P\phi N}{60}$ Volts.
Since Z conductors are divided into A parallel paths, conductors in series per path = Z/A:

$$\mathbf{E_g = (\frac{P\phi N}{60}) \cdot \frac{Z}{A} = \frac{P\phi ZN}{60A}}$$

B. DC Motor Back EMF (E_b) and Voltage Balance:

$$\mathbf{V = E_b + I_a R_a \Rightarrow I_a = \frac{V - E_b}{R_a}}$$

$$\mathbf{E_b = \frac{P\phi ZN}{60A}}$$

C. Electromagnetic Torque Equation:

Mechanical gross power developed equals electromagnetic power:

$$P_{\text{mech}} = T_e \cdot \omega = T_e (\frac{2\pi N}{60}) = E_b \cdot I_a$$

$$T_e\left(\frac{2\pi N}{60}\right) = \left(\frac{P\phi ZN}{60A}\right) I_a$$

$$\mathbf{T_e = \frac{1}{2\pi}\left(\frac{PZ}{A}\right)\phi I_a \approx 0.159\left(\frac{PZ}{A}\right)\phi I_a [N\cdot m]}$$

$$\mathbf{T_e \propto \phi \cdot I_a}$$

Problem:
A 220 V DC shunt motor has an armature resistance $R_a = 0.5\ \Omega$. When running at full load, it draws an armature current of 20 A. Calculate the Back EMF E_b.

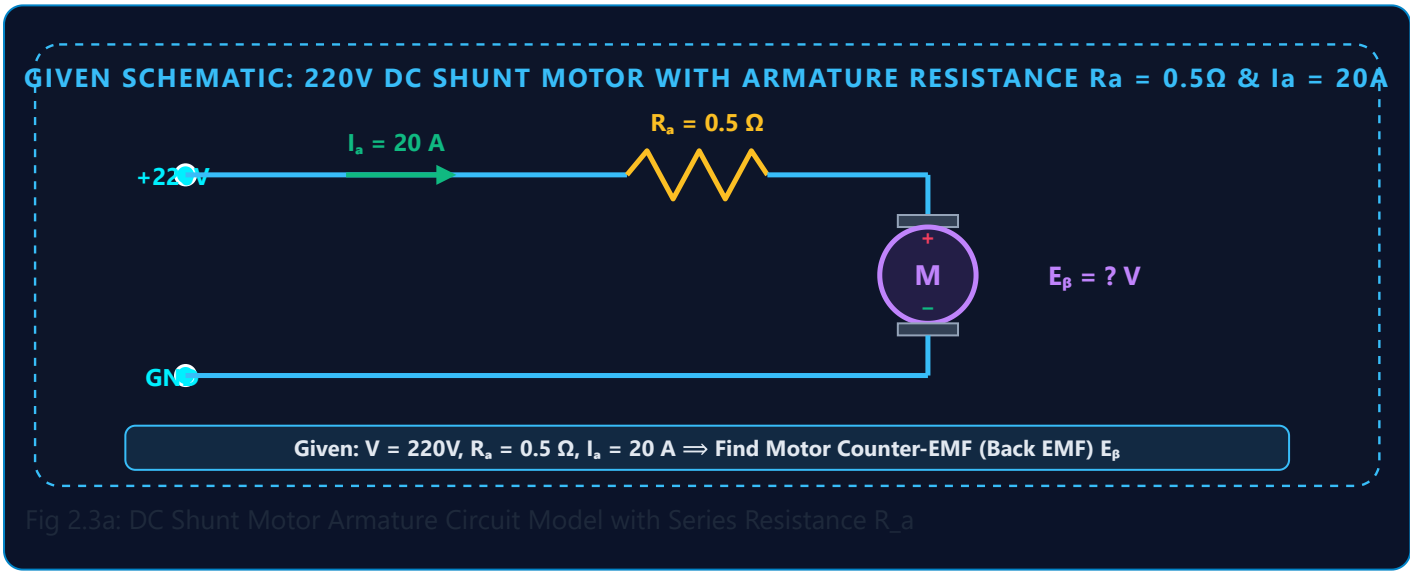


Fig 2.3a: DC Shunt Motor Armature Circuit Model with Series Resistance R_a

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Step 1 🚀: State Motor Voltage Equation

$$V = E_b + I_a R_a \Rightarrow E_b = V - I_a R_a$$

Step 2 ⚡: Substitute Values
$E_b = 220 - (20 \times 0.5) = 220 - 10 = \mathbf{210.0}$ Volts

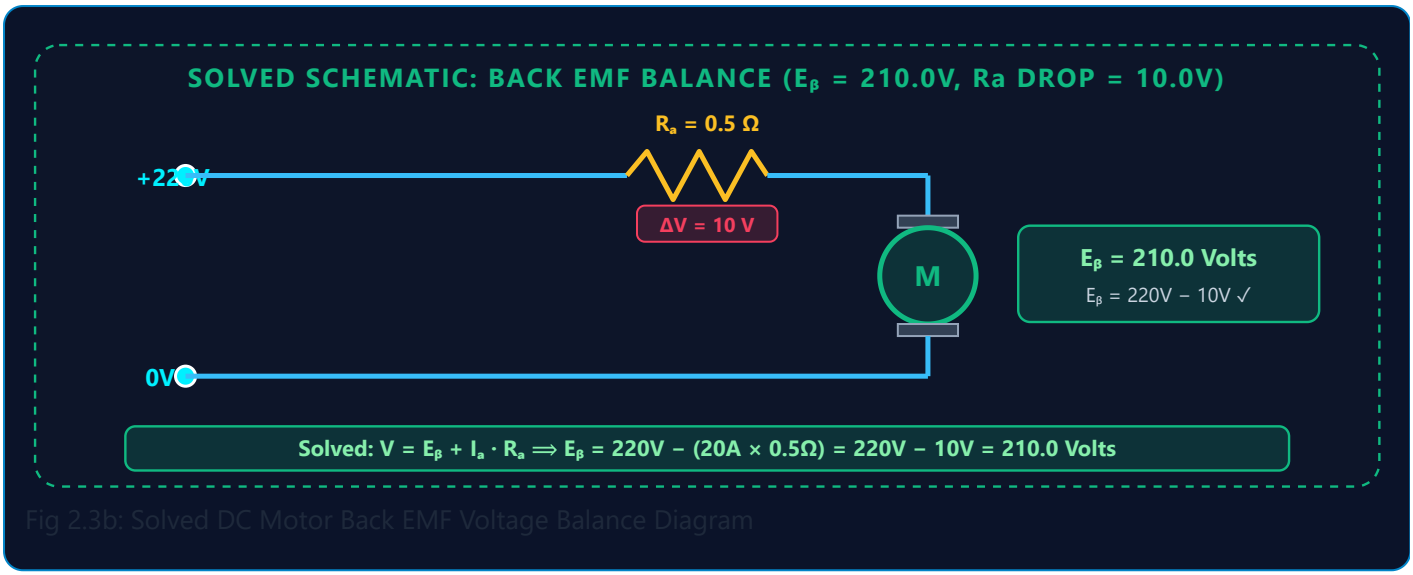


Fig 2.3b: Solved DC Motor Back EMF Voltage Balance Diagram

🎯 Back EMF generated: $\mathbf{E_b = 210.0}$ V.

University Question (8 Marks):

A 240 V DC shunt motor has armature resistance R_a = 0.4~\Omega and field resistance R_{sh} = 120~\Omega.

1. What is the armature current at the instant of starting (when N = 0)?

2. What is the total line current when running at rated speed with Eb = 228 V?

Step 1 🚀: Starting Armature Current (Eb = 0)

$I_{a,start}$ = (V - 0 / R_a) = (240 / 0.4) = 600.0,Amperes!

(Demonstrates why a 3-point starter resistor is mandatory!)

Step 2 ⚡: Running Armature Current

I_a = (V - Eb / R_a) = (240 - 228 / 0.4) = (12 / 0.4) = 30.0,Amperes

Step 3 🎯: Total Line Current IL = I_a + I_{sh}

I_{sh} = (V / R_{sh}) = \frac{240}{120} = 2.0 A

IL = 30.0 + 2.0 = 32.0,Amperes

🎯 Starting Current = 600.0 A (Dangerous without starter) | Normal Running Line Current = 32.0 A.

University Question (10 Marks):

A 230 V DC shunt motor runs at 1000, pm while taking an armature current of 40 A. The armature resistance is 0.25~\Omega.

If the mechanical torque is reduced by 50% and magnetic flux per pole remains constant, find the new operating speed.

Step 1 🚀: Initial Back EMF Eb_1

Eb_1 = V - I_{a1} R_a = 230 - (40 × 0.25) = 230 - 10 = 220.0 V

Step 2 ⚡: New Armature Current for 50% Torque

Since torque T \propto φ I_a and flux φ is constant:

T_2 = 0.5 T_1 ⇒ I_{a2} = 0.5 I_{a1} = 0.5 × 40 = 20.0 A

Step 3 🎯: New Back EMF Eb_2

$Eb_2 = V - I_{a2} R_a = 230 - (20 \times 0.25) = 230 - 5 = 225.0$ V

Step 4 🏁: Speed Relation (Eb \propto N)

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \implies N_2 = N_1 (\frac{E_{b2}}{E_{b1}}) = 1000 \cdot (\frac{225}{220}) = 1000 \cdot 1.0227 = 1022.7 \text{ rpm}$$

🎯 New Operating Speed: $\mathbf{N_2 = 1023}$ rpm (Speed rises slightly as load lightens).

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Basic Electrical & Electronics Engineering (`BELE-001`)

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 Question 4 (1 Mark): Hysteresis vs. Eddy Current Loss Frequency Dependence

Question:
How do core losses change if a 50 Hz transformer is operated on a 60 Hz supply at constant voltage?

> 📊 **Question Schematic / State Specification:**

```
Question Frequency Variation: V/f ratio changes flux density Bₘ ∝ V/f.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Loss Scaling Proportions:

        `Pₕ ∝ f · (V/f)¹.⁶ = V¹.⁶ f⁻⁰.⁶ ⇒ Decreases!`

        `Pₑ ∝ f² · (V/f)² = V² ⇒ Remains Constant!`
```

At constant voltage, flux density decreases as frequency rises. Hysteresis loss decreases slightly (proportional to $f^{-0.6}$) while eddy current loss remains constant, causing total core losses to decrease.

🎯 **Verified Result:** Question 4 step-by-step resolution per MMDU university marking scheme.

📘 Question 5 (1 Mark): Significance of Back EMF in DC Motors

Question:
Why does a DC motor draw dangerously high current if started without an external starter?

> 📊 **Question Schematic / State Specification:**

```
Question Starting Circuit: At N = 0 ⟹ Back EMF Eb = 0 ⟹ Armature Current Iₐ = V / Rₐ.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Current Surge Formula:

        `Iₐ,ₛₜₐᵣₜ = (V - 0) / Rₐ = 230  V / 0.5  Ω = 460  A ≈ 15–20 × Iᵣₐₜₑd`!
```

At standstill (N=0), back EMF is zero. Because armature resistance R_a is extremely small (~0.5 Ω), direct connection to rated voltage produces a destructive current surge that will burn the windings unless limited by a starter.

🎯 **Verified Result:** Question 5 step-by-step resolution per MMDU university marking scheme.

📘 Question 6 (1 Mark): Slip Condition of Induction Motor at Synchronous Speed

Question:
Why can an induction motor never run at synchronous speed (N = N_s)?

> 📊 **Question Schematic / State Specification:**

```
Question Relative Velocity: Rotor speed N vs Stator RMF speed Nₛ ⟹ Relative motion ΔN = Nₛ - N.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Induced EMF Zero Vector:

        `If N = Nₛ ⇒ Relative Speed = 0 ⇒ d&Phi;/dt = 0 ⇒ Eᵣ = 0 ⇒ Iᵣ = 0 ⇒ T = 0`!
```

If the rotor reached synchronous speed, rotor conductors would not cut any rotating magnetic flux. Induced rotor EMF, current, and electromagnetic torque would drop to zero, causing the rotor to immediately decelerate.

🎯 **Verified Result:** Question 6 step-by-step resolution per MMDU university marking scheme.

📘 Question 14 (2 Marks): Open-Circuit and Short-Circuit Test Diagnostic Roles

Question:
Explain why the wattmeter reading in an Open-Circuit (OC) test represents core loss only, while in a Short-Circuit (SC) test it represents copper loss only.

> 📊 **Question Schematic / State Specification:**

```
Question Diagnostic Regimes: OC Test [Rated Voltage, Negligible No-Load Current I₀] vs SC Test [Rated Current, Tiny Voltage Vₛc].
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Decoupled Loss Models:

        `OC Test: I₀ ≈ 2–5% × I_rated ⇒ I₀² Rc_u ≈ 0 ⇒ W₀ = P_iron`

        `SC Test: Vₛc ≈ 5–10% × V_rated ⇒ B_m ≈ 0 ⇒ P_iron ≈ 0 ⇒ Wₛc = Pc_u`
```

In the OC test, current is so low that copper loss is negligible. In the SC test, the applied voltage is tiny, so core flux and core loss are negligible, isolating the respective loss mechanisms perfectly.

🎯 **Verified Result:** Question 14 step-by-step resolution per MMDU university marking scheme.

📘 Question 18 (4 Marks): Transformer Efficiency and All-Day Efficiency

Question:
A 50 kVA distribution transformer has iron loss P_i = 500W and full-load copper loss Pc_u = 800W. Calculate: (1) Full-load efficiency at 0.8 pf lagging, (2) Load kVA at which maximum efficiency occurs.

> 📊 **Question Schematic / State Specification:**

```
Question Loss Energy Flow: Input Power ➔ [Iron Loss 500W + Copper Loss 800W] ➔ Output Power = kVA × pf.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Efficiency Formulations:

        `1. Full-Load Output = 50  kVA × 0.8 = 40  kW = 40,000  W`

        `Total Full-Load Losses = P_i + Pc_u = 500 + 800 = 1300  W`

        `Efficiency &eta;_(fl) = [ 40000 / (40000 + 1300) ] × 100 = \frac{40000}{41300} × 100 = 96.85%`

        `2. Load fraction for Maximum Efficiency: x = √{P_i / Pc_u,f_l} = √\frac{500}{800} = √(0.625) = 0.7906`

        `Load kVA for Max Efficiency = x × 50  kVA = 0.7906 × 50 = 39.53  kVA`
```

Maximum efficiency occurs when the transformer is loaded to **39.53 kVA** (79% of rated capacity), where variable copper loss equals 500W.

🎯 **Verified Result:** Question 18 step-by-step resolution per MMDU university marking scheme.

📘 Question 22 (6 Marks): Complete Theory and Speed Control of DC Shunt Motor

Question:

Derive the Torque equation of a DC Motor. Describe the two methods of speed control for a DC Shunt Motor (Armature Resistance Control and Field Flux Control) with relevant circuit diagrams, equations, and speed-torque characteristics.

> 📊 **Question Schematic / State Specification:**

```
Question Speed-Torque Characteristic Slopes: Rated speed N_rated ➔ Armature control drops speed below N_rated; Field control raises speed above N_rated.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Torque & Speed Derivations:

        `1. Electrical Power converted to Mechanical: P_m = Eb I_a = T · ω`

        `Substitute Eb = \frac{&Phi; Z N P}{60 A} and ω = 2πN / 60:`

        `T · \frac{2πN}{60} = [\frac{&Phi; Z N P}{60 A}] · I_a ⇒ T = [ \frac{P Z}{2π A} ] · &Phi; I_a = 0.159 &Phi; Z I_a (P/A)  N-m`

        `2. Speed Governing Relation: N = \frac{V - I_a R_a}{k &Phi;}`

        `Method A (Armature Resistance): R_ext added in series with R_a ⟹ N &darr; below rated speed. Wasteful I² R loss.`

        `Method B (Field Flux): Rf added in series with shunt field ⟹ If &darr; ⟹ &Phi; &darr; ⟹ N &uarr; above rated speed. Highly efficient!`
```

Field control provides a constant-horsepower drive above base speed, whereas armature control provides a constant-torque drive below base speed.

🎯 **Verified Result:** Question 22 step-by-step resolution per MMDU university marking scheme.
